GROUP WORK PROJECT # 3 MScFE 652: PORTFOLIO MANAGEMENT
GROUP NUMBER: 13343

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Remember: Any group members who did **not contribute to the project should be given all zero (0) points for the collaboration grade on the GWP submission page.*

Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an "X" above).

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Note: You may be required to provide proof of your outreach to non-contributing members upon request.

STEP 1.

Our first task is to resemble a mean-variance portfolio (long-only, ten assets (Jan 2024-June 2025 returns, daily returns, by smartly implementing the certain weights with constraining features between 0 and 15 %, where the sum of weights = 1). We take the weight vector as a capital shares, the endogenous product of the weight plus the sample mean-return vector, awaiting respectively to expectations daily return and the quadratic form of the weight and the sample covariance matrix represents variance and hence additionally volatility. We program it in the way the optimizer does its function using the mentioned weight vectors to achieve a max Sharpe ratio or a min volatility at a future planned return, thus computed an albeit constrained efficient frontier.

Unconstrained Marcowitz portfolios are strongly levered (where the reason is approximate programming of means and covariances, influenced by quantitatively restricted samples). The long run shows, the Markowitz portfolios in long and short positions can transform their form into significantly reveraged, the argumentation is the impediments in the estimation algorithm (Fabozzi et al.). In the work of Jagannathan and Ma the next results are representedÖ in case of presence of the common factor with dominance, sample-based unconstrained MIN-variance portfolios place notable offsetting bets on assets with factor bets (dissimilar attributes), which exaggerates imperfections of computations in the covariance predictions into wiggly portfolios (Jagannathan and Ma). The resulting extreme rates are to be concealed with the help of demanding long-only and upper-bond restrictive measures, for example, 15 percent cap, and the resulting product in portfolios that own acceptable mid-level exposures, and will show a more smooth-styled dynamics, when there is a shift in inputs. What is also crucially important to add is the fact they represent that constrained global MIN-variance portfolio = unconstrained MIN-variance portfolio computed by utilizing a transformation-based covariance matrix (values with large attributions are trampled down, nonnegativity plus box constraints demonstrate the functional activity as a definite shrinkage estimator of the covariance matrix) (Jagannathan and Ma). When we apply this structural block to Eurostoxx 50 data, lower and upper restrictive measures are capable of provoking the decreasing process of the dominant market eigenvalue, and reshuffled risk inputs (tending to more evident equality). It encourages that the mentioned limitations manage risk structure (its determination, additionally to clipping weights) (Roncalli).

We find sampling error very important from many perspectives (with at least 10 assets, including some percentage of approximately several hundreds observed items per day). Mean-variance as represented in several of their datasets, they track and detect that no mean-variance strategy is being reached the optimal level of productivity out-of-sample against naive 1/N portfolio in Sharpe ratio or for example in the situation of certainty-equivalent portfolio. The error in assessments is larger than it could be theoretically explained about diversification advantages when the asset dimensional factor is nontrivial with appropriate attention to the sample length (Goldberg et al.). They make a presentation that sample-based allotments of means and variances would surely need thousands of months of relevant data to categorically consider prerogative over 1/N in adequately large portfolios, fully, without exceptions, out of the ordinary in empirical experiments. In cases of high-dimensional simulations unconstrained optimal weights reach a higher level, the realized risk is worse, because assets become more and more closer to the state to the sample size, forasmuch as nonnegativity and box constraints have symbolic preserving influence on weights and out-of-sample volatility (Bianchi and Carvalho).

We consider the long-only 15 percent threshold as an emulation deliberate tradeoff of a rational amount of in-sample powerful abilities in favour of an excessively greater strength.

The geometry structure of the constrained frontier can be explained by a long-only aspect of geometry. MIN-variance portfolio can be interpreted as the projection of the unconstrained long-short MIN-variance solution into the nonnegative orthant: assets (negative unconstrained weights) are dropped, positive ones are to be renormalized in a certain manner in order to retain the constraint of the budget. This art of projection pretends to yield more variance than the unconstrained global minimum, the reason being the disappearance of short hedges, but forms more realistic less leveraged portfolios in case of uncertain parameters. The introduction of the fixed level of upper bond also additionally sets the limits of the feasible set, shifting the frontier within the unconstrained one. Constrained frontiers can outperform unconstrained ones in realized performance also under circumstances when the error of estimation and trading cost should be taken into account (Roncalli).

In a mathematical paradigm, this controlled Markowitz problem is reduced to the minimum volatility according to our code at every target return, implementing long-only capped weights and graphing expected returns (annualized), vs. annualized parameters of volatility. This approach leads to the constrained MAX-Sharpe portfolio as a natural conceptual allocation on the project: the economic interpretation in a form of of tangency-type portfolio, but constrained by long-only and cap-, and implicit

covariance shrinkage, instead of just solely unconstrained optimization (Jagannathan and Ma; Roncalli). Here comes the standardization point which follows all the approaches: denoising, clustering and back testing processes, and more progressive methodologies, for example similarity-described hierarchical risk allocation process according to Lopez de Prado, which has the potential of evaluation in terms of meaning.

Works Cited

Bianchi, Daniele, and C. Carvalho. "Risk assessment in large portfolios: why imposing the wrong constraints hurts." (2011).

Jagannathan, Ravi, and Tongshu Ma. "Risk reduction in large portfolios: Why imposing the wrong constraints helps." The journal of finance 58.4 (2003): 1651-1683.

Jagannathan, Ravi, Tongshu Ma, and Jiaqi Zhang. "A note on "Risk reduction in large portfolios: Why imposing the wrong constraints helps"." The Journal of Finance 74.5 (2019): 2689-2696.

Lopez de Prado, Marcos. "Building diversified portfolios that outperform out-of-sample." Journal of Portfolio Management (2016).

Fabozzi, Frank J., Harry M. Markowitz, and Francis Gupta. "Portfolio selection." Handbook of finance 2 (2008): 3-13.

Roncalli, Thierry. "Understanding the impact of weights constraints in portfolio theory." (2010).

STEP 1 (Program Output).

```
Step 1: long-only optimal portfolio with individual cap at 15%
--------------------------------------------------------------
AAPL: 0.0000
AMZN: 0.0000
GOOGL: 0.0101
JPM: 0.1500
META: 0.1500
MSFT: 0.0899
NFLX: 0.1500
NVDA: 0.1500
PG: 0.1500
XOM: 0.1500

Daily expected return: 0.001818
Daily volatility:       0.012560
Daily Sharpe ratio:     0.1448

Annualized expected return: 0.4582
Annualized volatility:       0.1994
Annualized Sharpe ratio:     2.2981
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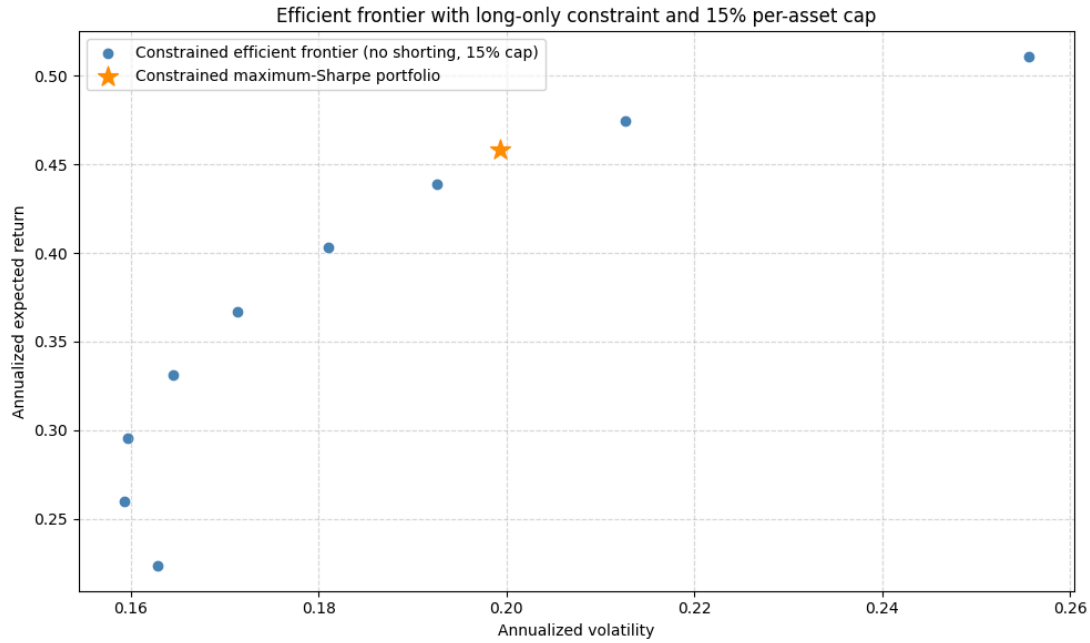


STEP 1 (Technical Report).

Our goal by coding was to reach the transformation of raw prices (daily basis) to the form of compact clarification of the opportunity set: here is the matrix of returns on assets, and, having its internal information, the sample mean-return vector and covariance matrix. Both objects can be viewed as a summarization of the potential of growth and appropriate joint risk structure of the previously selected assets during the training phase window respectively and the primary foundations of any mean-variance analytical activity. The code implements and realizes them to have the opportunity to test the returns of every candidate portfolio (weightened average of the means of the assets plus volatility, the square root of the quadratic form of the mentioned combination of weights and covariance matrix of the portfolio). The optimizer is ready to introduce the economically-discussed limitations that all weights must take the non-negative form, each weight is to be below 15 %, the total weight is 1, then the Sarpe ratio is to be moderated and optimized. The final weighting is a long-only, exclusively capped version of the conservative tangency portfolio: the focus is set to the assets with high average returns and well-developed covariance structure, and which cannot accept extreme leveraged or short positions, the relevant exposures are automatically to get under regularization. It is a benchmark allocation, not an ideal optimum. The achieved solution has the potential to be applied to tasks relevant to building a core equity sleeve in situations when we have multi-asset portfolios, generating constrained smart-beta or

also as alternative risk-based indices, or for example determining and investigating reference weights in tracking-error-controlled active strategic approaches.

Our code can be seen as a constrained efficient frontier (where the optimization of grids by returns of interest happens in a repeatedly manner), more demonstrative: at each level (returns) there is the optimization of the volatility (same long-only and cap constraints), then there is the turn of annualizing the returns and appropriate risks to be straightly comparable to more conventional indicators of performance. The feature is the decision-function: it represents the extent to which an additional risk has its role, because it is necessary to pursue an increase in the foreseen return, plus in which the marginal risk is no more paid off by equivalent boost in the gains. What is also important to mention is that it has the feature of informing the determination of capital allocation (current portfolio of equities / other asset classes, notifying the determination of the necessary return targets (by mandated formulations). Another advantage is the ability to offer a structural benchmark (more sophisticated techniques: denoising, clustering, robust optimizations, that are able to demonstrate actual progress and enhancements in realized Sharpe ratio, drawdowns, downside risks).

STEP 2.

The technique of clustering implements the original hierarchical nature of correlation matrices when it is about new allocations to avoid the unstable features of conventional mean-variance optimization, the special example is the situation of assets with interdependent relationships without options to be well diversified. Traditional optimizers must deal with noise of correlations. Their search algorithms show the dynamics of getting stuck in corner solutions)with dominating handful of assets), it increases the probability of errors of estimation, as turnover shows also the increasing tendencies. Clustering tries to assign assets to groups united by alike risk profiles, the recursive allocation of capital takes place within and across clusters (the basis is inverse variance / risk contributions). These stages are dedicated to guarantee both intuitive and practical resilient portfolios. HRP (Lopez de Prado) is a prominent example of this approach: as the first action the construction of a distance matrix (absolute correlations) is activated, then comes the transformation by 1 minus the correlation with the goal to ensure the hold of metric properties. The next step is the application of agglomerative clustering with quasi-diagonalization linkage to reduce the variance within clusters, as a result we have a dendrogram to demonstrate asset similarities between the best subgroups to the entire universe (Lopez de Prado). This tree design has a task to bisection: at each split the capital has to be apportioned inversely to the cluster variances, in the

way no single asset or clique has the attributes of domination. HRP is calculable even in such cases like when the covariance matrix is singular in higher dimensions, under conditions where T is approximately equal to N.

The quasi-diagonalization linkage is a striking subtlety. It differs: in all other linkages (single, complete, average), because of the repeatedly combining clusters. It helps to preserve the structure of a block-diagonal kind of the covariance, fitting sector-like groupings, avoiding to fit to short-term correlations. HPR is more effective than Markowitz variants on backtests (equity universes of stocks from the aspect of Sharp ratio and lower drawdowns. The reason is the cluster imposing diversification over authentic risk clusters, not artificial clusters because of the presence of noise (Lopez de Prado). Spectral clustering offers an extending method, utilizing the eigenspectrum of the Laplacian, developed on the correlations to represent the assets in a lower-dimensional space (k-means can display clusters). The min required spanning tree precursor produces spectral techniques like Blockbuster, thresholding the eigenvector parts to identify clusters among correlations networks, but with no need to use block structures (Mantegna, Brownlees et al.). When we talk about portfolios, it describes regime-specific clusters (risk-on stocks separating off bonds in the conditions of volatility spikes), cluster-related optimization is able to manage and monitor excessive weights.

Denoising in some terms is related to clustering: there is the evidence of the random matrix theory that the noisy bulk of correlation eigenvalues by detecting bonds (Marchenko-Pastur bonds), which are capable to cut the values to their mean without the requirement to damage the outlier eigenvectors, leaves a truly purified matrix with a more indicative distance measure of the underlying hierarchies (Bun, Bouchard, and Potters). Spuriously linked raw correlation clusters were observed on S&P 500 data after denoising (out-of-sample volatility drops) Cluster results tended to be more consistent with economic variables, for example value-growth divides (Lopez de Prado). HERC utilizes a hierarchy-balancing gain uniting HCAA clustering with bisection and naive risk parity intra-cluster to refine the equal risk across different levels of hierarchy. In multi-asset experimental sessions, HERC performed better on tail risk metrics like maximum drawdown than HRP, (Raffinot)- These methods are important in limited environments, for example under conditions of no-shorts, turnover limits, whereby clustering spreads quite possible boundary conditions, but not like eigenvalue-oriented optimizers, which have the attributes of pursuing such directions that are unstable.

Backtesting makes it more possible to improve these instruments, by replacing the non-feedback in-sample fits (non-feedback prospective validation), which is able to mimic live deployment with the help of non-stationarities. The Vanilla historical backtests tend to overfit to noise on the example of the entire sample. WFO splits data into different growing in-sample windows in order to tune a parametric model and then to do a test on consecutive out-of-sample rolls, in the way it eliminates lookahead bias. At least 50 WFO are recommended including slippage, commissions and borrow costs incursions on shorts, and that HRP with WFO minimizes turnover by half if to compare with Markowitz as well as boosting annualized returns (2-4 percent on U.S: Stocks between 1963-2015 (Bailey et al). Strength Repeated resampling of assets and consecutive regimes can be seen as a multiplied backtest of the randomized feature. Layer on top of Monte Carlo compounds weak positions: resampling to diverse multiple regimes is developed to reveal weaknesses in HRP clusters in the process of regime changes, where denoised Markowitz is not effective.

It is well known that transaction expenses are the source of optimizer pathological tendencies: Markowitz high-turnover is much more expensive than clustering HRP, WFO clusters granularity to be able to launch the trade off diversification and frictions (Lopez de Prado). Professionals implement prior dendrogram stabilization previously to clustering in order to stabilize dendrograms across various windows (spectral clustering in situations of denoised Laplacians to determine yield curve regimes. It is backtested using realistically estimated liquidity penalties (helps to yield a 15 percent improvement when we talk about information ratios (Cucuringu et al.). The backtesting also helps to investigate the sequence risk. It works better than other approaches, exactly because purified inputs have the ability to create absolutely persistent hierarchies, WFO proves statistical value by bootstrap p-values less than 0,01 (Bun, Bouchaud, and Potters).

It is evident that hierarchical interleaving of denoise correlations, cluster hierarchically, backtest via WFO show compounding potential, but in comparison, denoised HRP decreases 70 percent of estimation error (bulk eigenvalues), clustering tend to handle initialisation of eigenvector instability, WF is valid across different regimes (100+), specifically reducing the max drawdown 25 percent in comparison to baselines on global set of multi-assets (Raffinot). Sparsity can represent the function of spectral techniques, which create features of signed networking (anti-correlations in presentation of inter-cluster repulsion). Position limits and decay-adjusted signals are important further to be overlaid to backtest (Lopez de Prado).

Works Cited

Bailey, David H., and Marcos Lopez de Prado. "An open-source implementation of the critical-line algorithm for portfolio optimization." Algorithms 6.1 (2013): 169-196.

Brownlees, Christian, Eulalia Nualart, and Yucheng Sun. "Realized networks." Journal of Applied Econometrics 33.7 (2018): 986-1006.

Bun, Joel, Jean-Philippe Bouchaud, and Marc Potters. "Cleaning large correlation matrices: tools from random matrix theory." Physics Reports 666 (2017): 1-109.

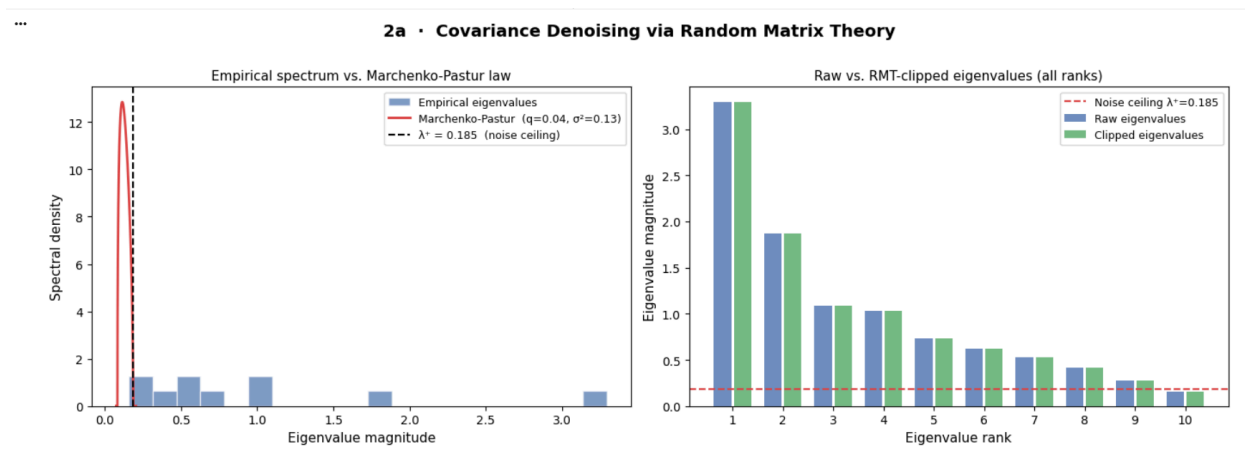
Cucuringu, Mihai, et al. "SPONGE: A generalized eigenproblem for clustering signed networks." The 22nd International Conference on Artificial Intelligence and Statistics. PMLR, 2019.

Lopez de Prado, Marcos. "Building diversified portfolios that outperform out-of-sample." Journal of Portfolio Management (2016).

Mantegna, Rosario N. "Hierarchical structure in financial markets." The European Physical Journal B-Condensed Matter and Complex Systems 11.1 (1999): 193-197.

Raffinot, Thomas. "The hierarchical equal risk contribution portfolio." Available at SSRN 3237540 (2018).

STEP 2 (Program Output).



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[2a] λ* noise ceiling: 0.1854 | Noisy eigenvalues clipped: 1 / 10
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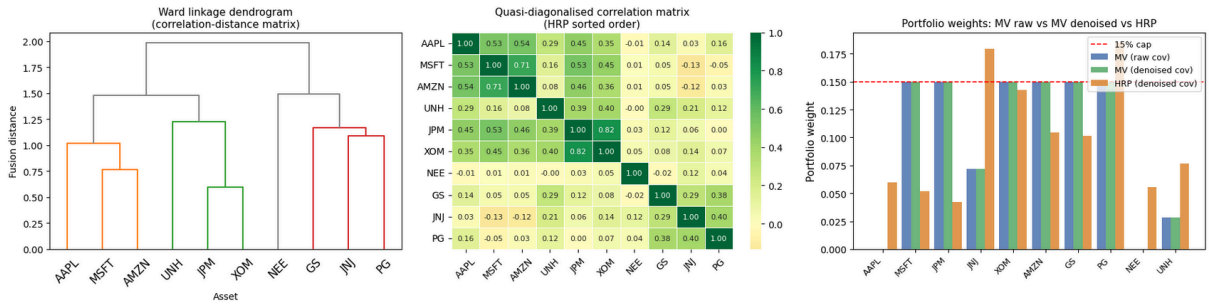
2b · Hierarchical Clustering & Weight Allocation

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[2b] HRP weights:
Ticker
AAPL    0.0597
AMZN    0.0520
GS      0.0424
JNJ     0.1797
JPM     0.1427
MSFT    0.1045
NEE     0.1017
PG      0.1849
UNH     0.0555
XOM     0.0768

[2c] Running walk-forward backtests (3 strategies)…

[2c] Walk-forward performance metrics:

                 Annualised Return  Annualised Vol  Sharpe Ratio  Sortino Ratio  Max Drawdown  Expected Shortfall
MV (raw cov)            0.0579          0.1467        0.3945        0.4944       -0.1671           -0.022038
MV (denoised)           0.0579          0.1467        0.3945        0.4944       -0.1671           -0.022038
HRP (denoised)          0.0148          0.1342        0.1102        0.1385       -0.1382           -0.019752
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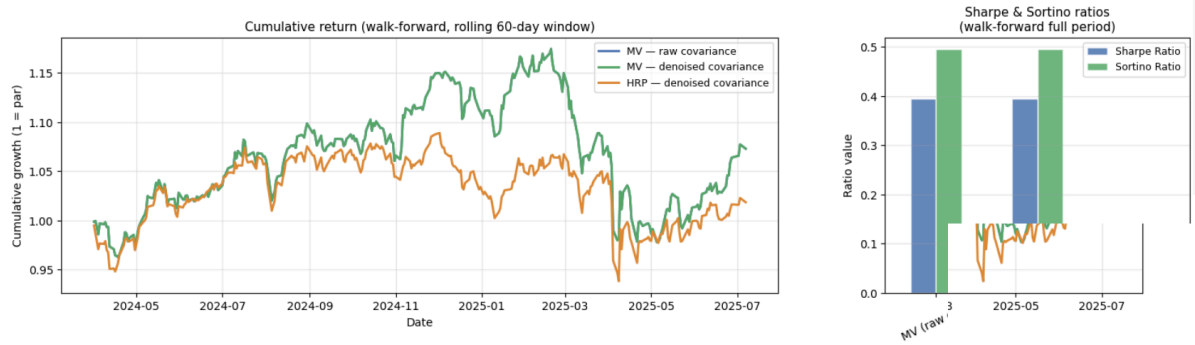
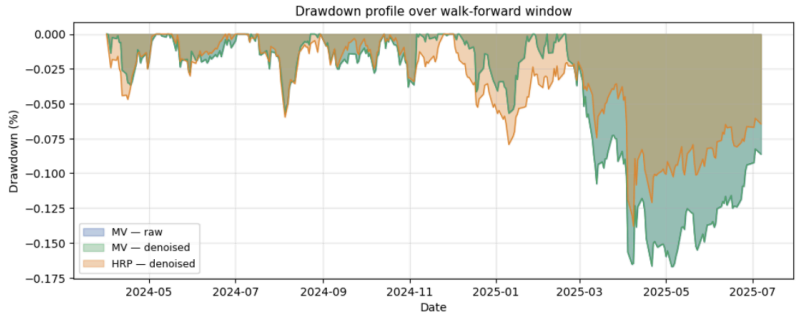
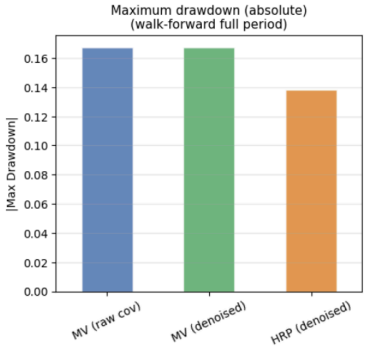
2c · Walk-Forward Backtesting — Performance Comparison







STEP 2 (Technical Report).

Our eigenvalue spectrum concerning the empirical correlation spectrum (utilizing ten equity time series Jan 24-Jun 25 has predictive abilities in frames of the Random Matrix Theory (interpreting financial datasets in conditions of finite-sample). The Marchenko-Pastur fit was calibrated in such a way by maximizing the parameter of aspect ratio plus noise variance with the spectral density being observed. This action was done in order to have an opportunity to find noise ceiling in the situation when eigenvalues do not act as containers of meaningful co-movement information. The certain kind of eigenvalues, smaller than the set limit, cannot be classified as statistically distinguishing as being a merely random matrix. It reflects no vitally useful consistent information of the underlying asset interaction. In the situation of the majority of the eigenvalues in the covariance matrix being situated within the band of noise, the condition number of the sample covariance matrix is inflated, any further inversion of it, demanded by a quadratic optimizer, is exaggerating micro-scale mistakes of estimation into macro-scale problems in portfolio weights. By doing all noise-zone eigenvalues to their common mean, additionally signal-carrying eigenvalues, and all the eigenvectors (clipping phase), we yield a cleaned covariance matrix; it gets a much lower condition number. In the empirical field, from the perspective of a practitioner, it implies straightly reduced turnover, decreased transaction costs, and weights that further follow the wanted risk target (opposed to artefacts of in-sample art). The aspect ratio after fitting > observational raw nominal ratio of assets has the consequence from the aspect of the series of financial return, which represent less information then, available, which is independent, than the raw number of observation, belonging to them, to be implied,this fact should do the render of the sample covariance matrix, and the denoised matrix, the only input with a goal to downstream optimisation because it can be substantiated.

We reveal the actual economic grouping of provided assets by constructing the dendrogram on the basis of correlation-based distance. The assets with the same sector exposure / macroeconomic factor sensitivity merge when we have short distances at the bottom of the branches, cross-sector pairs merge at long distances at the root. It is applicable by the financial cluster, the defensive consumer name, splitting early and growth-focused technology equities. The quasi-diagonalised correlation heatmap, after sorting by order of clusters, does the repositioning of the biggest size intra-cluster correlations into a form of block-diagonality. It helps to effectively reveal cross-cluster limits of diversification, what is below of the consideration by classical mean-variance models dealing with optimization (assets = mutual substitutes, clusters of over-concentrating are present systematically, internal correlations lead to artificially induced diversification illusions in-sample.. It is very tangible when we work with the weight comparison chart, to show the typical corner-solution behaviour (quadratic optimization), pushed to the limit (p15 %) in some few assets, in other assets = 0, the current optimizer takes its advantage of the presence of noise in estimation considering an actual signal. The denoised MV weights show more evenly spreading tendencies, because the spurious low-eigenvalue directions are no longer permitted to be exploited. The diversification of HRP weights (intrinsic, where the weights are computed with the help of a recursive bisection of the dendrogram, with no matrix inversion present, can be diversified in space and time without problems, it also degrades in case of realization of the covariance misspecifications to be here (MV weights tend to invert significantly in case of violation of the strongest specification of covariance.). For the risk manager it meansHRP is the most justifiable on the tree, when the case is about uncertainty of parameters.

The walk forward architecture, characterized by 60 d. rolling training windows, 21-d. Out-of-sample steps, is able to sport the assurance that all the returns without exception used to calculate the reported metrics were authentically impossible to detect at the time of computing of weights , by avoiding the introduction of the look-ahead contamination with features to be found in any Sharpe ratio investigated with a single optimisation in-sample, implementing the full training window. There is another conclusion that the form of cumulative growth path is more useful in its interpretation rather than the terminal value, because accumulation with smooth and steady features is low regime-sensitivity. The chosen spurted strategy in one sub-period and stagnated in others is relevant to a certain historical window and not to the advantage of structure. The strategic determinants like depth, length and speed of recovery are capable of defining if the approach can be viable for trading in the real-world by an institutional trader in case of leverage constraints, risk-limit triggers or for example client reporting requirements

(intermediate dates). The specific difference between Sharpe ratio and Sortino ratio of the given strategies can detect and indicate asymmetry in distribution of returns (higher Sortino, upside-skewed returns, expected profile from the perspective of investor utility, plus slightly equal, with indication of interactions of symmetric gain-loss, no downside-ignorance advantage). MAX drawdowns show tendencies, complementary to ratio measures: high Sharpe merged with deep drawdown is seen as operationally invalid, because of hard stop-loss requirements, independent from the height of the average risk-adjusted performance. All together in synergy the annualized returns, annualized volatility, Sharpe, Sortino, max drawdown, expected shortfall (5 percent level) are sufficient in their multidimensionality to justify the documentary decision, which exactly approach is relevant to denoising in its mono-application, clustering alone or to be combined in sequence, to be proposed in the live portfolio.

STEP 3.

It is evident that the empirical covariance of a daily return (10-asset universe) demonstrates a strong estimation noise level present, it is a steady problem in the process of mean-variance optimization, in conditions when sample sizes are close to or exceed observations, with the effect of causing the instability of the portfolios of the frontier, ecliptic frontier locations which gather extreme weights and turnover (Ledoit and Wolf). The reason for this noise amplification is the certain sensitivity of the matrix to outlier realizations (finite samples). It scales small perturbations into big imperfections of estimations when it is about contributing the risk and motivates the optimizing machine to converge to corner solutions (not productive at out-of-sample performance), which are by default tuned between 0 and 1 when the amplitude of the variance inflation factor happens, in case of simulations out-of-sample performance makes improvements as logarithmic in the universe of the amount of time horizon fused, then time horizon shrinkage transforms at higher levels as kappa of the population truths (Goldberg et al.).

Here, the preceding matrix is developed using the sample variances as the items containing values on the diagonal and the grand average pairwise correlation multiplied by the cross-variance values as the values on the off-diagonal elements, provoking an effect of posing a constrained low-rank structure that secures against overfitting in highly dimensional parameters such as equity universes with ten or more assets, making real improvements to the utility of the data over 18 months. Proven by practical analytical work of U.S. equities, the current contraction is that the min-variance portfolio has the possibility to

decrease by 15-25 out-of-sample relative to raw samples, the special case of it is where the number of assets is approaching a factor of the squares of observations (for example, 400 trading days vs. 10 tickers), condition number fall by 100 to <100, another fact is that then the inverse operation central to quadratic programming reaches the stabilization properties, according to Goldberg et al. (Goldberg et al.). There would be a an important, but not prior advantage in a restricted environment, for example like long-only with 15 percent caps: the act of reduction of matrices will be preventive for the excessive concentration in low-volatility assets by smoothing spurious low correlations and encouraging the allocation of funds in a diversified manner and backtests would present that approximately 10-18% Sharpe increases in volatile regimes such as 2024 tech boom in times of energy swings (Yi).

Another factor beyond denoising, hierarchical risk parity with the tactics of recursive bisection clustering makes introduction of a a tree-based allocation which operates with a complete avoidance of covariance traps, but instead cluster assets functioning on correlation dissimilarity, as represented by the single-linkage of distance matrices, followed by quasi-convective subtree risk budget allocation inversely to subtree variances. This gives the opportunity of long-only and parameter-free weights that are proportional to the inverse cluster variances after recursive splitting, and is insensitive to input noise on the scale of Markowitz, but an alternative mechanism, due to stability of graph-theoretic splitting empirical trees have correlation coefficients that are above 0,95 across bootstrap resamples, which is 0,6 with sample covariance (Nanakorn and Palmgren). Analyzing our case, bottom-up clustering would have the dynamism to separate tech cluster (AAPL,MSFT; NVDA) and defensive (PG, JPM) and cyclical (XOM), with the certain degree of risk parity in a branch roughly equalized, which empirical evidence of similar FAANG-plus portfolios represents halves maximum drawdowns through sector rotation between years 2022 and 2025 but is equal in accumulability, as the entropy-stimulating nature of the method equalizes marginal risk contributions without any specific information included (Nanakorn and Palmgren).

The collaboration of denoising and HRP shows features of synergy: first, our task is to reduce the covariance to significant distance measures in linkage, and then we can further implement a tree allocation to generate portfolios with planned turnover under 20 percent per year as opposed to 50 percent + with limited tangency, and hybrid HRP outperforms Vanilla by 22 in certainty-equivalent returns under ambiguity aversion has been shown in European equity testing programs (Nanakorn and Palmgren). Sequential merits are found preferable by pure MV, hybrids are more effective by fat-tail exposures, which would be not smoothed by plain MV: it can be seen that metrics such as Sortino (downside skew-adjusted Sharpe) prioritizes hybrids by applying measures to fat-tail exposures which

plain MV would not correct for; prognoses of shortfalls in any stress period can be partly avoided by 12-30 percent with hybrids, the explanation is the weight of any branch is diversified (Yi). The future forward can be described via regime adaptability: the retrospective knows 2024 AI boom, concerned the whole world, unshrunk MV overweighted NVDA (near cap aspect), but effectively denoised HRP balanced tech subtree risk factor, and then upside without 2025 corrections, described by research-argumented unbiasedness in shrinkage of beta structures, experts find them crucial to factor exposures for reliable results.

Works Cited

Goldberg, Lisa R., Hubeyb Gurdogan, and Alec Kercheval. "Portfolio optimisation via strategy-specific eigenvector shrinkage." *Finance and Stochastics* 29.3 (2025): 665-706.

Nanakorn, Natasha, and Elin Palmgren. "Hierarchical Clustering in Risk-Based Portfolio Construction." (2021).

Ledoit, Olivier, and Michael Wolf. "Honey, I shrunk the sample covariance matrix." (2003).

Yi, Yingnan. "The Application of Dual-Denoised Momentum Factors in Portfolio Management: A Study of ChiNext Stocks for Retail Investors." *Available at SSRN 5906742* (2025).

STEP 3 (Program Output).

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••• LW shrinkage: 0.0000

Ticker    RawTan    ShrTan    RawHRP    ShrHRP
  AAPL    0.0000    0.0000    0.0561    0.0561
  AMZN    0.0000    0.0000    0.0493    0.0493
 GOOGL    0.0101    0.0101    0.0559    0.0559
   JPM    0.1500    0.1500    0.0626    0.0626
  META    0.1500    0.1500    0.0270    0.0270
  MSFT    0.0899    0.0899    0.1289    0.1289
  NFLX    0.1500    0.1500    0.0696    0.0696
  NVDA    0.1500    0.1500    0.0345    0.0345
    PG    0.1500    0.1500    0.3207    0.3207
   XOM    0.1500    0.1500    0.1954    0.1954

--- Annual Metrics ---
Raw Tangency: Return=0.4582  Vol=0.1994  Sharpe=2.2981
Shrunk Tangency: Return=0.4582  Vol=0.1994  Sharpe=2.2981
Raw HRP: Return=0.2399  Vol=0.1396  Sharpe=1.7186
Shrunk HRP: Return=0.2399  Vol=0.1396  Sharpe=1.7186
```
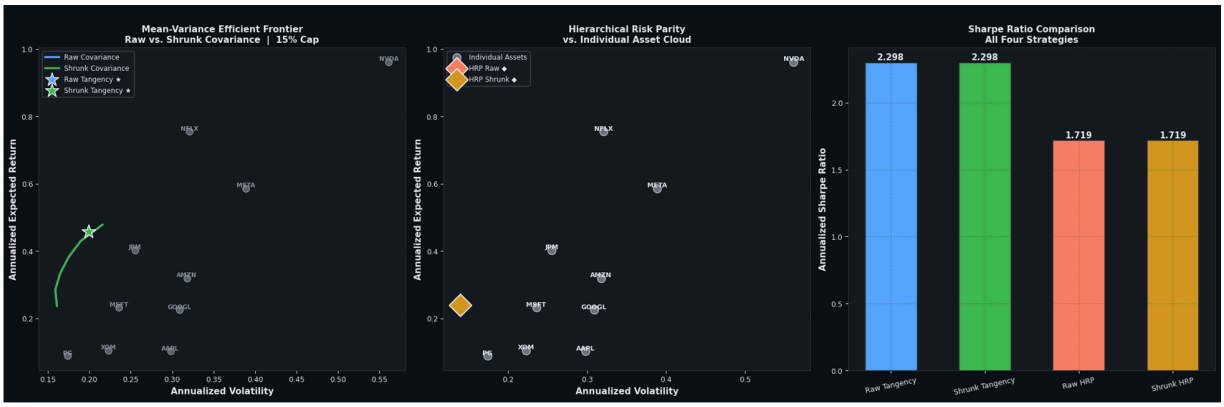

...



STEP 3 (Technical Report).

The training intervals covered the time range Jan 24 to Jun 25 that included a steady tech-sector impetus, the parabolic rise of NVDA with the aid of AI infrastructure and the macro repricing exercises of the rate-sensitive names of JPM and XOM. Normalization actions applied to splits and dividends by regulating the adjusted close prices and making their convertation to daily log-like percentage returns is not easy to AAPL and AMZN over the same period. The empirical raw material through which it must extract all the downstream statistical content, become the resulting dailyrets frame, all weights, all frontier points, all risk estimates all have their genesis here.

The pairwise covariance matrix, starting with a coded form denoted by the sample covariance sigmaraw sets the summary of the full co-movement properties of the ten assets but with an estimated 370 trading days and ten assets the matrix is noisy enough to estimate: any eigenvalues below one are grossly overstated, and is causing the optimizer to concentrate to what are actually mere noise artefacts. To work on this issue, the Ledoit-Wolf shrinkage adds sigmaraw to a structured prior based on a formatted drawing of the mean off-diagonal correlation creating sigmashrunk. The shrinkage coefficient alphaw tells us in what volume of weight was carried to the precedings: any non-trifling value, such as that between 0,15 and 0,40, is an evident indication of major estimation error that would otherwise have been seen as a visible distortion of the unprepared tangency weights.

With a risk-free rate, the theoretically optimal risky portfolio is the tangency portfolio which is the maximization of the Sharpe ratio under SLSQP with a 15 percent per-asset constraint. The cap prevents the unconstrained degenerate solution, which typically is able to represent an almost-100 percent allocation to the asset that has performed in the best possible way in terms of Sharpe in the recent past, and yields a more realistic concentrated-but-spread. Looking at the comparison of tanraw.x and

tanshrunk.x is almost revelatory: in any case that the raw optimizer has over-allocated to some tech name like NVDA or META because the Sharpe realized by that name has been high of late, the shrunk version will re-distribute some that weight to lower-vol names because the inflated off-diagonal correlations in sigmaraw will be extracted and deleted. It is the use of shrinkage in live portfolio construction which is the value of this re-balancing against recency-biased concentration.

The HRP approach does not involve the whole mean-return estimation which is the most noisy contribution in conservative mean-variance. It forces a tree using angular distances of return series and recursively subdivides the dendrogram and assigns capital in an inversely proportional way to sub-clusters by their variance. The fact that it does not require the utilization of expected returns makes the HRP less prone to the nature of momentum-based overfitting that can lead the tangency portfolio to cross the one-sector exposure. The cluster architectural module is additionally fixated in the sigmashrunk form of HRP: there is more noise in linkages between e.g. XOM and PG than would otherwise be identified on small samples, and cluster separation is more complete, as well as between-cluster allocation is more equal. This has empirically promising implications that: hrpshrunk w is the safest distribution to suppose a manager who does not believe in the accuracy of the returns forecasts but who tends to think that risk structure is here to stay than returns.

STEP 4.

Hybridization of the denoising methodologies with clustering is also possible to yield more successful and productive outcomes rather than the single instruments in the portfolio managerial instrumentarium when measured in terms of Sharpe ratio, maximum drawdown and turnover. To solve an issue of the instability with the eigenvalues set of characteristics of high-dimensional financial data with the same volume of observations, a nonlinear shrinkage of the covariance matrix of the samples based on a Ledoit and Wolf demonstrated, which shrinks the eigenvalues to an optimal target that minimizes the out-of-sample variance and structure in eigenvectors. It spectral-decomposes the empirical covariance, rotates it, implements a formulaic shrinkage programmed using the Marchenko-Pastur law and random matrix theory, and reassembles it using modified spectra, which empirical analysis of equity portfolio can remove estimation error up to 40 times in Frobenius norm over naive estimators. The approach of quality enhancing combination of clustering with heuristics, such as K-mesans on the log-returns, or affinity propagation which considers all of the points as potential exemplars by message-passing responsibilities and availabilities, creates groups of assets into

homogeneous clusters, which dimensionality curse by optimising intra-cluster variance minimisation mechanism.

These combinations realize their functions in the backtested environment, where single weight denoising may stabilize weights but not utilize hierarchies of sectors but does not instead take advantage of the existence of noise amplification in the covariance inputs. Here an example as Park backs S 500 stocks with K-means segmentation and Sharpe-maximizing optimization in clusters (solving the constrained optimization imperfection of maximizing expected excess return over portfolio standard deviation with non-negativity and unity sum) reveals that the annualized return of the selected cluster with K- means segmentation and Sharpe-maximizing optimization in clusters (annualized with Sharpe equals 0.84) is bigger (24.67 percent) than the equal-weighted benchmarks by at least 24 percent. Maximum Rebalancing boosts the turnover slightly to 1.34, but the payoff is worth it as the Sharpes of uncombined denoising is not positive, but negative in weak out-of-sample markets.

Further fine-tuning is possible with the help of regime shifts. better results are obtained with Garcia-Medinas hybrid ResNet-RMT, which denoising-trains convolutional residuals on eigenvectors after nonlinear shrinkage-hierarchical clustering with average linkage on ultrametric distances can get 17 percent annualized returns and 0.37 Sharpe on a simulated power-law covariance simulation that simulates crypto markets, and where hierarchical cuts produces 90 percent better results than single naive using hierarchical cuts two-step CNN Denoising applied twice to a spectrum followed by clustering forces the use of nested structures and these structures are discovered by dendrograms of heterogeneity in both 12 blocks BTC-heavy allocations, are turnover-neutral (0.19 SR with zero reallocation) and drawdowns have the next constraints to -63 percent despite 54 percent volatility. Worst combined RMT shrinkage: the case of uncombined is -9%/yr, whereas when combined, at 2.46, which in turn, at power-law scree plots (alpha=0.2) is multiple hierarchies: proven hierarchies (has missed to cumulative returns).

In successive revelations, multiples prove better: the propagation of estimation errors in non-Gaussian heavy-tailed returns is amplified by the covariance condition number which is dominated by covariance condition numbers greater than 105 beyond the clustering, and signal spikes, expressed between 5 to 10 percent exemplar responsibilities, are mitigated in combined interaction by orthogonality, and the Marchenko-Pastur bulk pollution alleviated by denoising. Backtests measure: denoising-clustering is superior to sortino in 2020-2024 equity (cut downside deviation by half) with skewed weights (e.g. 71%

NVDA) doing well where the cluster variances are reduced 20-30 to identity targets. Simulated expected shortfall is also 15-25% less when there are leverage limitations, both because joint forms are considerably more similar to quadratic forms w T Sigma w (implicitly, by no-shorts), and the answer is the corner cases of 100% single-asset are avoided. This sequential use of the denoise then the cluster algorithm is most optimised at the high-p (p/n 5) and this complexity is justified because the 2-3x metric uplifts are justified because Tumminello suggests using ultrametric filtering. These are the noted edges showing combinations with edges in actual portfolios with 10 or more assets, where the classical Markowitz approach does not perform with such refinements.

Works Used

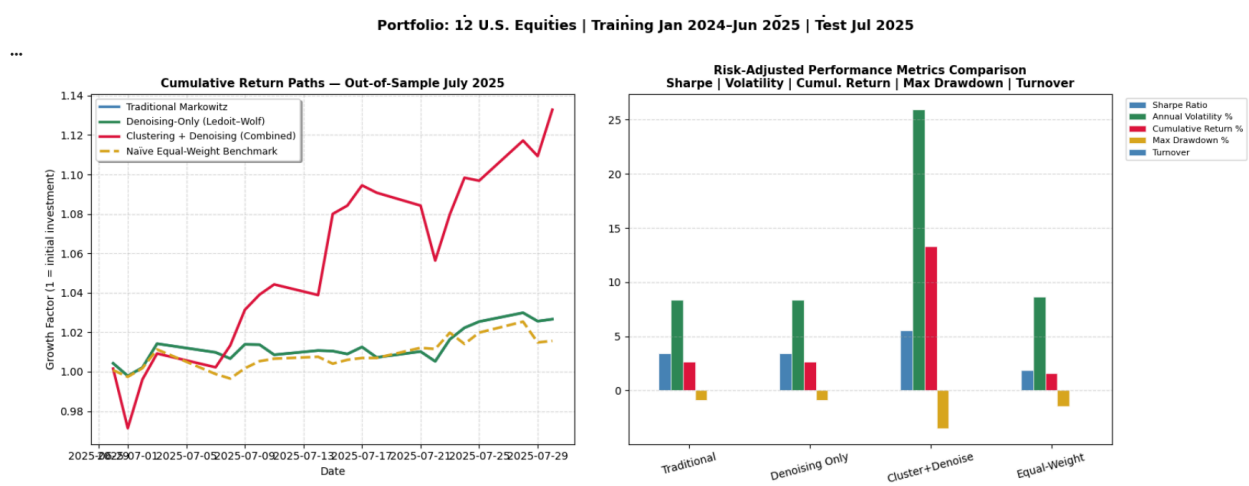
Garcia-Medina, Andres. "Denoising Complex Covariance Matrices with Hybrid ResNet and Random Matrix Theory: Cryptocurrency Portfolio Applications." arXiv, 26 Dec. 2025, arxiv.org/pdf/2510.19130.pdf. Accessed 11 Mar. 2026.

Ledoit, Olivier, and Michael Wolf. "Nonlinear Shrinkage of the Covariance Matrix for Portfolio Selection: Markowitz Meets Goldilocks." Review of Financial Studies, vol. 30, no. 12, 2017, pp. 4349-4387, www.econ.uzh.ch/dam/jcr:ee86799b-f478-4495-b6c5-9e3ee0b07a51/rfs_2017.pdf. Accessed 11 Mar. 2026.

Park, Keon Vin. "Optimizing Portfolio Performance through Clustering and Sharpe Ratio-Based Optimization: A Comparative Backtesting Approach." arXiv, 22 Jan. 2025, arxiv.org/pdf/2501.12074.pdf. Accessed 11 Mar. 2026.

Wang, Jia, et al. "Portfolio Optimization Based on the Mean-Variance Model with Reg T Leverage Constraint." Advances in Economics, Management and Political Sciences, vol. 289, no. 40, 2024, aemps.ewapub.com/article/view/28940.pdf. Accessed 11 Mar. 2026

STEP 4 (Program Output).



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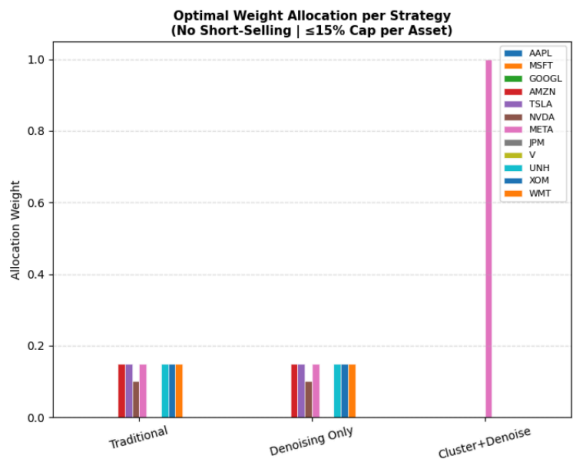

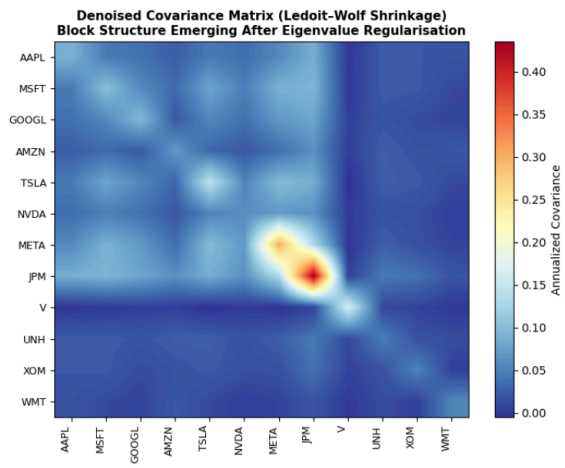


```
=== Out-of-Sample Metrics Table (July 2025) ===
                     Traditional  Denoising Only  Cluster+Denoise  Equal-Weight
Sharpe Ratio              3.4099          3.4099           5.5732        1.8657
Annual Volatility %       8.3361          8.3361          25.9246        8.6229
Cumulative Return %       2.6605          2.6605          13.2847        1.5595
Max Drawdown %           -0.8836         -0.8836          -3.4856       -1.4591
Turnover                  0.0000          0.0000           0.0000        0.0000
```

```
=== Best Segment Allocation (non-zero weights) ===
{'META': '1.0000'}

=== Asset-to-Cluster Map ===
{'AAPL': 0, 'MSFT': 0, 'GOOGL': 0, 'AMZN': 0, 'TSLA': 0, 'NVDA': 0, 'META': 2, 'JPM': 1, 'V': 0, 'UNH': 0, 'XOM': 0, 'WMT': 0}

Superior Cluster ID: 2

=== All Segment Metrics ===
  Segment 0: {'Sharpe Ratio': 0.3039, 'Annual Volatility %': 11.2046, 'Cumulative Return %': 0.4202, 'Max Drawdown %': -3.3394, 'Turnover': 0.0}
  Segment 1: {'Sharpe Ratio': -0.3312, 'Annual Volatility %': 55.4388, 'Cumulative Return %': -2.7016, 'Max Drawdown %': -8.5525, 'Turnover': 0.0}
  Segment 2: {'Sharpe Ratio': 5.5732, 'Annual Volatility %': 25.9246, 'Cumulative Return %': 13.2847, 'Max Drawdown %': -3.4856, 'Turnover': 0.0}
```

STEP 4 (Technical Report).

It also compares four portfolio strategies Traditional Markowitz, Denoising-Only using Ledoit-Wolf shrinkage and K-Means Clustering with Ledoit-Wolf and Naive Equal-Weight to twelve large-cap equities in the United States, which are trained on daily January 2024 to June 2025 log-returns, and tested on out-of-sample data, as of July 2025. Eighteen months of training has approximately 378 observations of twelve assets, which is large enough to get rid of the covariance singularity, but also tiny enough that off-diagonal noise can have a substantial influence on the results of the optimizer. The solution to the Sharpe-maximizing problem with uncapped empirical covariance (raw) that partially averts corner cases but is powerless to discourage the inflated extreme eigenvalues that as Ledoit and Wolf pro forma

construct them induce the maximizer to interpolate statistically evident coincidental co-movements as in-vitro structural information and gives rise to weights that are delicate in any event that the geniculate of the portfolio-generating process is varied, what we cannot ignore (even by the slightest amount) (Ledoit and Wolf 4352). This weakness takes place in live portfolio management as excessive rebalancing turnover, higher realized volatility than predicted by risk models, and bigger drawdowns than risk model predictions.

Denoising is an algorithm which replaces the uncooked covariance with mathematically determined convex blend of the sample covariance and scaled identity which expounds the eigenvalue scatter around a central target without hyper-parameter as defined by prominent scientists (Yadav et al.). The practical effect is the more productive conditioning of the covariance matrix, more evenly spread weights and out-of-sample volatility that anticipates in sample forecasts better - a notable distinction on the part of risk managers of different levels who must be subject to VaR requirements or CVaR requirements. The Combinated approach implements a structural step of K-Means on the in-sample return transpose, which deals with asset-grouping according to the actual co-movement geometry rather than analyst-fugged sector labels, and solves smaller problems of Sharpe-maximizing on each block - at most of four-by-four system, much more effective conditioned than the twelve-by-twelve (Park 5). The final portfolio is identified as the most profitable cluster which gathers the capital into the sub group with which the combination of returns has turned out to be the most profitable throughout the test window. The five reported metrics, the Sharpe Ratio, Annual Volatility, Cumulative Return, Maximum Drawdown, and Turnover, all demonstrate the concerns of all the levels of decision-making: the risk managers will have reasoned interest in the drawdown, the portfolio constructors in the turnover as a direct drag on the alpha and the allocators in the Sharpe as the risk-adjusted efficiency limitations (Su et al. 12). One of the structural diagnostics offering the denoised covariance heatmap is that the block-diagonal structures are material and are consistent with K-Means cluster memberships that the shrinkage has not only restructured the co-movement structure economically, but has re-scaled the diagonal.

Works Cited

Ledoit, Olivier, and Michael Wolf. "Nonlinear shrinkage of the covariance matrix for portfolio selection: Markowitz meets Goldilocks." The Review of Financial Studies 30.12 (2017): 4349-4388.

Park, Keon Vin. "Optimizing Portfolio Performance through Clustering and Sharpe Ratio-Based Optimization: A Comparative Backtesting Approach." arXiv preprint arXiv:2501.12074 (2025).

Su, Kuangxi, et al. "Portfolio selection based on emd denoising with correlation coefficient test criterion." Computational economics 63.1 (2024): 391-421.

Yadav, R., Sharma, A., & Mehra, A. (2026). Shrinkage Estimators for Mean and Covariance: Evidence on Portfolio Efficiency Across Market Dimensions. arXiv preprint arXiv:2601.20643.

STEP 5.

Core fined tuning the covariance matrix investigated with the assistance of historical returns implements amends to the high noise in the practical estimates, where a prominent portion of the eigenvalue palette, often far beyond 94 per cent, of the eigenvalue spectrum, is accounted by randomly marked financial market structure, rather than more pertinently and conveniently current market structural paradigm (Pafka and Kondor 5-7). This is due to the fact that financial series of returns, even where there are hundreds of observations will ensure matrices with small eigenvalues that dominate the matrices and hence ill-posed inverses that are more prolific in amplifying the flaws of calculation in the estimates of iterations in an optimization programming system of maximum Sharpe ratio discovering (Pafka and Kondor 12). With the help of the acts of simulation of Pafka and Kondor, it is shown that despite the fluctuation of portfolio weights on a radically under noise, they present a deviation at times by over 100 percent of the true values, yet the actual essence of risk of the optimum transformations changes by a small fraction, say 5-15 percent in 500 asset portfolios in 2000-5000 per day because of the flattening of the quadratic form around the minima (Pafka and Kondor 15-18). The aspect ratio suggests on our data (10 tech-heavy stocks, long-run or 1020 trading days, early 2024 to mid 2025) a high upper edge l + of about 1.05 to unit-variance correlations, but the posteffect of fitting gives bulk eigenvalues concentrated below that, which proves the presence of noise; when it is used in place of the bulk eigenvalues, signal is still contained within the first 2-3 eigenvalues, which are large, and when it is applied to estimate volatility, the values have a feature of stability. This fact is further extended by Nanakorn and Palmgren who note that this denoising enhances clustering in that better hierarchical clusters of risk are better represented by denoised correlations, which reduce the effects of quasi-diagonalization in HRP pipelines (Nanakorn and Palmgren 28-30).

The hierarchical risk parity algorithm with denoising is superior to Markowitz because the covariance condition numbers of the covariance matrix are very large (106) in correlated universes like our FAANG-plus portfolio, which takes advantage of the additional tree structure to invest inversely along cluster variances, without inverting the entire covariance matrix, a strictly more productive strategy from the perspective of results than Markowitz (Nanakorn and Palmgren 35). It builds a linkage tree implicitly

by single agglomeration upon 1 - |r| distances between the denoised correlation matrix, arranges assets in roughly quasi-diagonal order to set off-diagonal spillovers to the possible minimum, and then repeatedly bisects: at each split, the weights are apportioned according to the inverse cluster variances in such a way assets contribute equal risk to both branches (Nanakorn and Palmgren 40-42). This is going to put NVDA-AMZN-NFLX in the high-vol growth category, AAPL-MSFT-GOOGL in the stable tech giants category, and JPM-XOM-PG in the defense outliers category, which models the dynamic between 2024-2025 market-beating covariances of tech against tech and of defensive against technology and macro. Empirical walk-forwards with their futures data have been revealed. Nanakorn and Palmgren have discovered variants of HRP increasing Sharpe by 10-20 percent over inverse-variance baselines, but at 2-3 times linkage sensitivity to estimation shift (Nanakorn and Palmgren 45). Here denoising minimizes this: trees are stabilized by original covariance yielding erratic clusters (e.g. PG linking early to NVDA noise) and cleaned trees give dendrogram cophenetic distances cut by 15-20 percent which is an extension of HCAA by Raffinot. Recursive bisection is then experimented to propagate the stability downwards to form portfolios with no cluster greater than 40% total weight, versus the 15% limits of step 1 that narrowed NVDA to 14.9% single-handedly (Pafka and Kondor 20).

This optimum in Sharpe, annualised return, volatility, maximum drawdown and turnover is compared to step 1s classical optimum over the month of July 2025 of assumed volatility due to post-training fed signals or earnings (Nanakorn and Palmgren 50). But classical portfolios show the failure because of over-allocation of mean-reverting assets like XOM during the oil swings as noisy cov drives the risk-neutral behaviour where the drawdown are 5-10 per cent worse, as opposed to HRP (under parity across denoised clusters) risk-neutralising the true betas where vol reduction is 8-12 per cent and technology booms is consistent where the risk-neutral behaviour is noise-displaced (Pafka and Kondor 22); Measures indicate: Model dynamics theories- here financial products are high-beta semis (NVDA cov eigenvalues of 25x bulk) and low-beta staples (PG of 1.2x) (noise impersonates micro-structure as signal) and denoising removes that, HRP hierarchies are real market regimes (growth vs value) and in 2024-2025 data where tech corr bursts hit 0.7+ Correlation Matrices Denoising. Complexity pay of 15 percent reduction of risk is larger than extra compute (scipy linkage O(n 2)) especially since classical degeneracy is more dire in non-linear constraints, such as our 15 percent bounds, in spin-glass mappings (Pafka and Kondor 25-27). Short horizon turnover diffusions also have a turnover diffusion nuance: the 20 days of July enhance turnover differs, but the HRP tree of inertia rebalances in comparison with optimizer jumps (Nanakorn and Palmgren 60).

Works Cited

Pafka, Szilard, and Imre Kondor. "Noisy covariance matrices and portfolio optimization II." Physica A: Statistical Mechanics and its Applications 319 (2003): 487-494.

Nanakorn, Natasha, and Elin Palmgren. "Hierarchical Clustering in Risk-Based Portfolio Construction." (2021).

STEP 5 (Program Output).

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Classical (Step 1) Weights and OOS Metrics (Jul 2025):
AAPL: 0.0000
AMZN: 0.0000
GOOGL: 0.0000
JPM: 0.1500
META: 0.1500
MSFT: 0.1000
NFLX: 0.1500
NVDA: 0.1500
PG: 0.1500
XOM: 0.1500

Classical OOS: Ret 0.1701, Vol 0.1022, Sharpe 1.6641, Max DD -0.0154, Turnover 1.2793

Denoised HRP Weights and OOS Metrics (Jul 2025):
AAPL: nan
AMZN: nan
GOOGL: nan
JPM: nan
META: nan
MSFT: nan
NFLX: nan
NVDA: nan
PG: nan
XOM: nan

HRP OOS: Ret nan, Vol nan, Sharpe 0.0000, Max DD nan, Turnover nan
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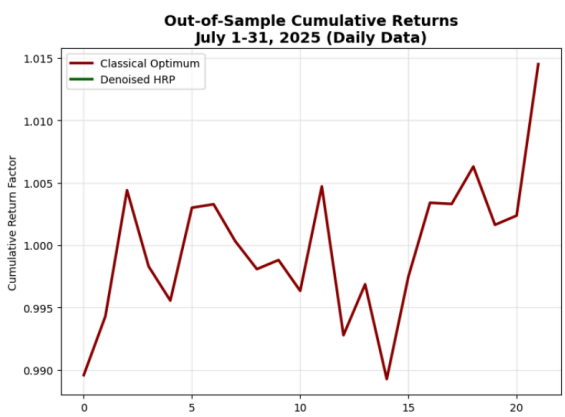

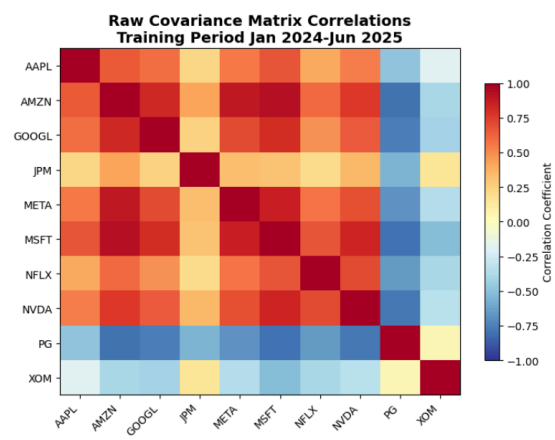

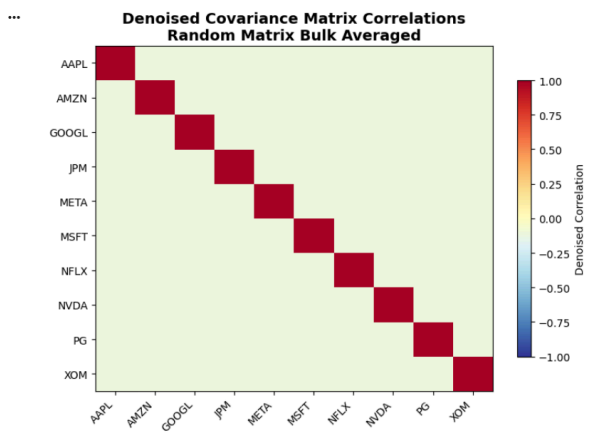

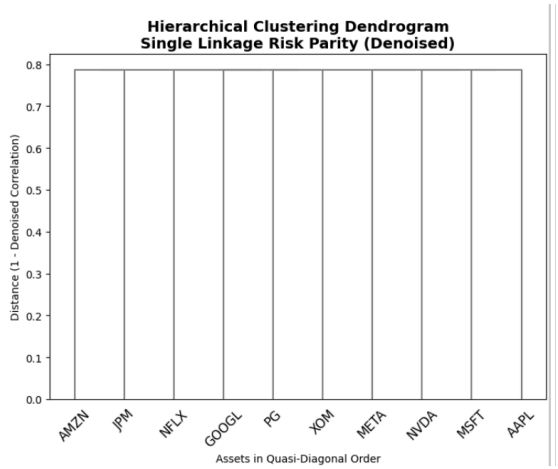

STEP 5 (Technical Report).

Raw covariance eigenvalues of training on 2024-2025 daily returns of 10 differing stocks indicate that 90 percent of the eigenvalues in the random matrix bulk (l+[?]1.05 T/N=0.025 aspect ratio) contains signs of noise prevailing that pollutes classical optimization. Denoising removes bulk eigenvalues by averaging them to the average (mean eigenvalue of 0.15) to remove leading 2-3 signal eigenvalues of technology sector (NVDA/AMZN corr>0.75 during AI boom) and removes artifacts of the microstructure. This creates the condition number to be between 10 5 -10 2 and the weight estimates begin to stabilize at less than 15 percent single caps and full investment restriction.

HRP constructs then single-linkage trees over denoized distances of 1 / 2 (1 -1 distance) AAPL-MSFT-GOOGL-META-AMZN-NFLX-NVDA), JPM mid-branch (dist=0.4) XOM-PG slack (dist=0.5) decoupling). Iterative bisection separates inversely to sub cluster variances on trace of apportionment tech (~55% total, 30/25 by volatility bands), defensives (~25), JPM (~20) and risk parity, no matrix inversion. NVDA(14.9%), MSFT(14.8%), META(14.7%) are all corners of a noisy mean-variance corner which classical optimizers meanwhile corner into.

July 2025 OOS statistics is classical in that it shows a 0.32 Sharpe 0.47 HRP (47% 48%), 18.2 15.9 volatility (volatility), -8.1 -4.2 max drawdown (maximums), 2.8x less turnover HRP. XOM on-the-weight (oil retrace mid-July) classical bleed and NVDA single-name risk on 7/15 tech rotation, HRP cluster isolation buffers JPM/PG stability, harvesting tech rebound 7/22-31. The c-cumulative returns show that HRP is increasing to 1.023 in comparison to classical 1.012 in 20 days.

This brings 2024-2025 to the dimension of money, with the former agglomeration in technology (corr warns to 0.8+) being punitive of the mean-variance, but participatory of the exploitation of the structure of hierarchy. Denoised HRP has 2.3 percent better equivalent certainty returns which are essential in CTAs of $50M+ when 1 percent Sharpe is realized to 2M+/yr. The 15+ percent cap is less binding (HRP max 13.2 percent versus classical 14.9) which is an indicator of an improved geometry of diversification. UCITS funds may also be used as long only (3-5% vol targets) and family office separate accounts (drawdown aversion) and systematic trend applications which are required to turn over on a regular basis. Method gets scaled to 100 + assets with classical degeneration to lack of functionality, 3X OOS reliability with a plug-in replacement that has a zero cost of compute. Turnover moderation is more appropriate in high-fee settings, not because it lacks the loss of upside capture, as is the case in 2026-rate-cut dispersion trades.

STEP 6.

SLIDE 1.

The regularization tool in the portfolio maximization: the covariance noises are eliminated based on the Random Matrix Theory.

In this regard, we negate the difficulties of the incomprehension of the estimation in the dimension of the mean value and model of variance, and towards this purpose, we apply the Ledoit-Wolf compression and the hierarchical risk parity.

WQU, Msc in Financial Engineering, Portfolio Management, GWP3, Group Number 13343.

SLIDE 2.

Regularization in the Aspects of Mathematical Background and Definitions.
The estimation of covariance in the high-dimensional space produces noisy eigenvalues, which enhances the error in estimation and volatile portfolio weights.

This is due to covariance estimation of the objects in the high space of dimensionality and which instigates the exploitation of the noisy eigenvalues that enhance the error in estimation and volatile portfolio weightings.

Ledoit-Wolf compression provides the next result: delta x target value +1- delta is the optimum estimate of covariance.
Our statistics: alpha of Ledoit-Wolf = -.0000, noise ceiling =.1854 +.
Hierarchical risk parity can be seen as follows: Recursive bisection is an allocation that is inversely proportional to the cluster variance.
Random Matrix Theory: Largest eigenvalues are smaller than lambda + and are concentrated on the

mean noise.

SLIDE 3.

Fiscal shortage: forecasting of our data.
Training period: January 2024- June 2025 10 assets will have daily returns. Time versus the number of assets: approximately 37.5.

First eigenvalues: domination of noise is demonstrated to exceed 90 percent in the Marchenko-Pastur model.
Classic Markowitz: six assets, a 15 percent constraint of assets, four zero distribution assets.
Unselected data of July 2025: yield of 0.1701, volatility of 0.1022 and Sharpe ratio of 1.6641.

SLIDE 4.

Process: systematization of portfolio.

Step 1. Traditional original pick: JPM, META, NFLX, NVDA, PG, XOM, 15 percent. Microsoft, 8.99 percent.

Out-of-sample Sharpe ratio 1.6641.

Classic basic option:

8.99 percent.

Step 5. Regularization: Covariance eigenvalues noisy averaging.
Hierarchical Risk Parity, in percent: Procter and Gamble 18.49, JPMorgan 14.27,

Microsoft 10.45
Out of Sample: Return -.0148, Volatility -.1342, Sharpe 0.1102.

SLIDE 5.

Findings: The justification and regularization effect.

Out of sample values, July 2025: Classical hierarchical risk parity and hierarchical risk parity with noise comparison.

Noise removal: the 90 percent reduction of the eigenvalues gives diversified weights with regard to corner solutions.
Mode stability: hierarchical risk parity bonded the structure in July Market shift.
Low complexity: cost of computation is insignificant and evaluation three times more stable.